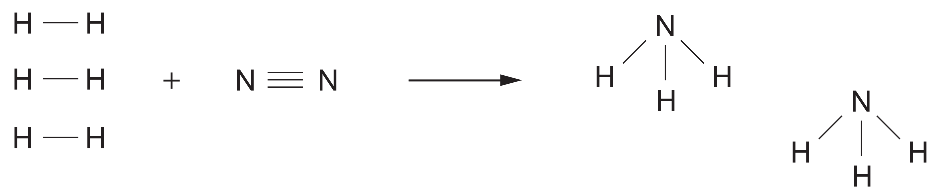
Examiner only

6. Ammonia is manufactured from hydrogen and nitrogen in the Haber Process.

(a) The equation shows the bonds which are broken and the bonds which are formed during the manufacture of ammonia.



hydrogen nitrogen ammonia

The relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
N≡N	945
H—H	436
N—H	391

(i) Calculate the **total** energy needed to break **all** the bonds in the hydrogen molecules and the nitrogen molecule. [2]

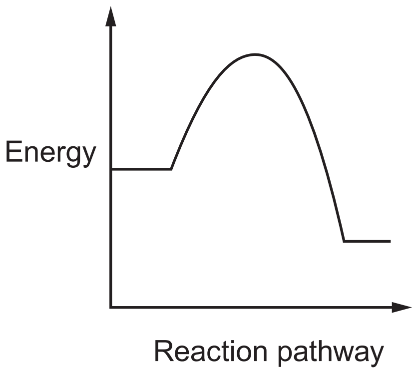
Energy = ... kJ

(ii) Calculate the **total** energy released when **all** the bonds in the ammonia molecules are formed. [2]

Energy = ... kJ

(iii) Use your answers to parts (i) and (ii) to calculate the overall energy change. [1]

Energy change = ... kJ *(b)* The manufacture of ammonia is an exothermic reaction. Put a tick (✓) in the box next to the energy profile diagram that shows an exothermic reaction. [1]



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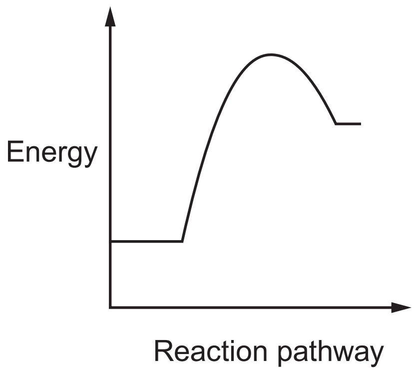

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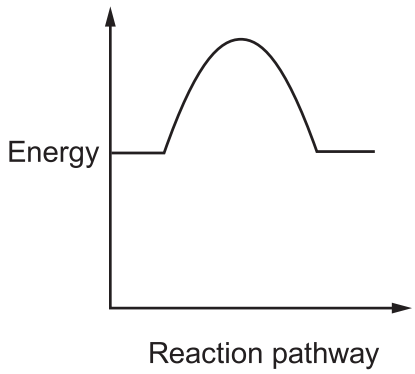

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(c) The table shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)				
	100	200	300	400	500
	Yield of ammonia (%)				
100	96.7	81.7	52.5	25.2	10.6
200	98.4	89.0	66.7	40.0	18.3
400	99.4	94.6	79.7	55.4	31.9

Use the information in the table to answer parts (i) and (ii).

(i) State what happens to the yield of ammonia as the temperature increases. [1]

....

(ii) One manufacturer carries out the Haber Process at 200 atm and 450 °C.

<u>Underline</u> the approximate percentage yield of ammonia formed under these conditions. [1]

10 % **30 %** **40 %** **58 %**

Examiner only

(d) Ammonia is used in the manufacture of nitrogenous fertilisers. One example of a nitrogenous fertiliser is ammonium nitrate. Ammonium nitrate is formed by the reaction between an acid and ammonia.

(i) Complete the **word** equation by naming the acid used in this reaction. [1]

ammonia + ... ⟶ ammonium nitrate

(ii) When ammonium sulfate solution is warmed with sodium hydroxide solution a pungent gas is formed. Damp red litmus paper is used to test this gas.

I. Describe the change in colour of the litmus paper during the test. [1]

....

II. State the property of this gas which causes the colour change. [1]

....

III. Name the gas formed. [1]

....

(iii) Nitrogenous fertilisers pollute streams and rivers. State how nitrogenous fertilisers get into these waterways. [1]

....

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